# MARCELINO SANDRONI DIAS

## Senior Software & AI Engineer

Engenharia que conecta sistemas distribuídos, produto e resultado financeiro.

**Location:** Fortaleza, CE (Fully Remote)
**Email:** marcelino.sandroni@gmail.com
**WhatsApp:** +55 11 91446-1993
**LinkedIn:** linkedin.com/in/marcelinosandroni
**Git:** github.com/marcelinosandroni
**Portfolio:** marcelinosandroni.com

## PROFESSIONAL SUMMARY

**A genuine full-stack engineer,** pairing strategic vision and business analysis with end-to-end delivery across UI/UX, mobile, frontend, backend, DevOps, data engineering and AI. Almost eight years in software engineering and fifteen in financial governance, building scalable solutions and critical systems that move billions of reais.

## KEY PRODUCTION IMPACT METRICS

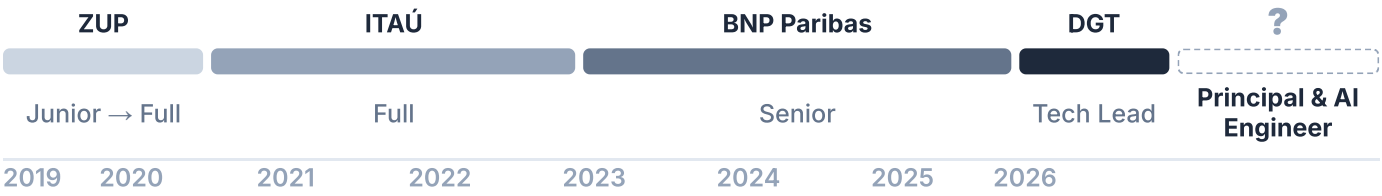
**+R$ 500Bn**
**ASSET MANAGEMENT**
Critical applications for fund custody.

**D+0 (Real-Time)**
**SYNCHRONOUS SETTLEMENT**
Real-time payment processing system.

**100M+ msgs/dia**
**HIGH SCALE**
Architecture for asynchronous event processing.

**R$ 24M/ano (ROI)**
**ARTIFICIAL INTELLIGENCE**
Contract saved through state-of-the-art AI implementation.

## CAREER TIMELINE IN ENGINEERING




## TECHNICAL SKILLS & SOFTWARE ARCHITECTURE

**Frontend & UI:** React, Next.js, Angular, Vue.js, React Native, Flutter, TypeScript, JavaScript, Micro-frontends, SSR, Storybook, Tailwind CSS, Design Systems.

**Backend & Frameworks:** Java (Spring Boot), C# (.NET Core), Node.js (NestJS), Go, Python, RESTful APIs, gRPC, WebSockets, Clean Architecture, DDD, CQRS, SOLID.

**Data & Messaging:** Oracle, PostgreSQL, SQL Server, MySQL, MongoDB, Redis, ClickHouse, Databricks, Apache Kafka, RabbitMQ, Event Sourcing, Circuit Breakers.

**Cloud, DevOps & SRE:** AWS, GCP, Azure, Oracle Cloud, IBM Cloud, Docker, Kubernetes (Helm), Terraform, CI/CD, OpenTelemetry, Prometheus, Grafana, Jaeger, FinOps.

**AI & Governance:** PyTorch, TensorFlow, YOLOv8, LangChain, MLflow, RAG, Agentes LLM, Visão Computacional, TDD, BDD, Testes E2E, SonarQube, SOX, IFRS/CPC.

## ACADEMIC EDUCATION & SPECIALIZATIONS

**BSc in Computer Engineering** — UNIVESP, 2021 – 2026

**BSc in Accounting** — Estácio, completed 2017

## LANGUAGES

**Spanish:** Básico — reading technical material and written communication.

**Professional English:** Fluent — meetings, negotiation, architecture and code review.

## PROFESSIONAL EXPERIENCE

**Senior Software Engineer & Tech Lead** | DGT — **(1 year) · Oct/2025 – Sep/2026**

**Stack:** Java, Spring Boot, Spring AI, Angular, TypeScript, Micro-frontends, PostgreSQL, Redis, gRPC, K8s, AWS.

- **Technical Leadership & AI Integration:** Led an 8-person squad through the restructuring of critical systems processing 100M+ messages/day. Corporate AI wired into the backend through Spring AI for real-time agentic flows, protecting a R$ 24M/year contract.
- **10x Faster Processing:** Migrated a bottlenecked relational architecture to Java microservices, with JVM tuning (ZGC) and low latency over gRPC and RabbitMQ.
- **Enterprise Micro-frontends:** Complex web interfaces in Angular, with reactive state (RxJS/NgRx), keeping the experience smooth under heavy load.

**Senior Software Engineer** | BNP Paribas — **(2 years and 10 months) · Dec/2022 – Sep/2025**

**Stack:** Java (Spring Boot/Batch), Angular, Micro-frontends, Oracle SQL, Kafka, OAuth2, K8s, Jenkins, Helm.

- **Bank Monolith Modernisation:** Technical pillar of a 12-engineer tribe, decoupling legacy systems (COBOL, VB) into event-driven architecture in Java with modular Angular interfaces (95% test coverage).
- **D+0 Settlement & Real Time:** Replaced D+1 pipelines with real-time fund reconciliation on Apache Kafka and Spring Batch, lifting assets under management by 45% with a 100% uptime SLA.
- **Global Cost and Orchestration:** Automated institutional ETF orchestration across B3 and international exchanges, with an estimated R$ 50 million in new inflows.

**Full Software Engineer** | ITAÚ — **(2 years and 5 months) · Jul/2020 – Nov/2022**

**Stack:** Java (Spring WebFlux), Angular, TypeScript, PostgreSQL, Redis, Kafka, AWS, K8s, Terraform.

- **Strategic Institutional Custody (+R$ 100 Bn):** Co-led a 15-engineer squad designing and running a B2B platform connected to B3, on cloud-native AWS with reactive Java APIs and Angular dashboards.
- **Crisis Management and Resilience:** Resolved Kafka concurrency bottlenecks under heavy volatility in under 24 hours, enforcing strict transactional idempotency.
- **8x Quality Improvement:** Technical mentoring and E2E automation (CI/CD), shielding the codebase from critical regressions.

**Full Software Engineer** | ZUP — **(1 year and 4 months) · Mar/2019 – Jun/2020**

**Stack:** Java, Angular, TypeScript, PostgreSQL, Redis, RabbitMQ, AWS, Docker, Kubernetes, CI/CD.

- **Rapid Progression and Cloud:** Promoted from Junior to Full within one year. An agile team of six engineers refactoring Java monoliths onto event-driven architecture, and migrated to AWS in half the agreed schedule.
- **Transactional Resilience:** Rebuilt asynchronous messaging reliability on RabbitMQ with circuit breakers and dead-letter queues, reaching 100% data accuracy.

**Accounting Analyst & Finance Manager** | Consultoria — **(15 years and 6 months) · Jul/2003 – Dec/2018**

- **Corporate Engineering Edge:** Team leadership and process redesign that cut the accounting close by 83%. Deep expertise in finance and tax rules, translated directly into software teams.